DMDCicero User Guide

General notes

Cicero, written by Aviv Keshet and freely distributed under the GNU General Public License as published by the Free Software Foundation and designed to act as a user interface or a ”client” for Atticus, which takes care of controlling the National Intruments output hardware. DMDCicero is a slightly altered version of Cicero, designed to add a simple communications channel between the Cicero client and Morse (from version 1.20.7 onwards), a client software for controlling ALP 4.2 -based digital micromirror devices (DMDs) manufactured by Vialux.

This document will describe features unique to DMDCicero. For instructions on how to use Cicero, see the Github page https://github.com/akeshet/Cicero-Word-Generator .

Using DMDCicero

DMDCicero works exactly like Cicero, except for an extra feature allowing for simple communication between DMDCicero and Morse. The communication between DMDCicero and Morse is handled through the text file *\\storage.yale.edu\home\navon-627019-FASPHY\Li Lab\DMD\Runtime\DMDruntime.txt*.The first row of the text file tells Morse whether to reload all uploaded sequences to the queue of the DMD, and the second row changes their refresh rate.

Controlling Morse via DMDCicero takes place via the *Variables* tab of the Cicero interface. Each time a sequence is run, DMDCicero will look for two variables: *DMDfreq* and *DMDupdt*. If variables with these names are found on the variable list, DMDCicero will notify the user via the runtime window textbox. The variable values control DMDCicero as follows:

- *DMDupdt*: If the value of this variable is 1, Morse will add all uploaded sequences to its queue in the order they are listed in the sequence list. If this value is anything else, Morse will not add the sequences to the DMD sequence queue.
- *DMDfreq*: If *DMDupdt* equals 1, Cicero will update the refresh rate of the uploaded sequences while adding them to the queue. The updated refresh rate corresponds to the value of *DMDfreq* in Hz. Valid range: 0.1-22,727 Hz.

Additionally, DMDCicero saves the runlogs to the shared drive, under *\\storage.yale.edu\home\navon-627019-FASPHY\Li Lab\DMDCicero Runlogs\YYYY\MM\DD*

Changes in v.1.20.7.9

The main change was the added functionality to run serial port devices (*Baud*=115200, *StopBits*=1, *DataBits*=8 – this was tested on SRS SG384 signal generator) connected to Cicero computer via the RS232 tab. This feature supports raw data commands (text only) + parameter commands (text+numeric variable). The current version supports two simultaneously connected RS232 devices.

To set up the devices, one needs to:

1) Create one or two dummy RS232 devices (these should show as *unassigned* in the channel manager) with logical index 0 (for the first device - Dev1) and 1 (for the second device - Dev2).

2) These devices are assumed to be attached to ports COM3 (Dev1) and COM4 (Dev2) on the Cicero computer, but these ports can be changed via the variable tab by creating variables named *RSport1* (Dev1) and *RSport2* (Dev2), whose values control the COM port number.

3) Once the devices are properly set up, one can assign command during a sequence via assigning the wanted RS232 groups into the sequence. Cicero will try to output the commands in the beginning of the timestep the commands are assigned to.